

Crab Nebula Pulsar Wind 26D UQFF Supernova Remnant

Daniel T. Murphy

PAPER_766: Crab Nebula Pulsar Wind 26D UQFF Supernova Remnant

Author: Daniel T. Murphy
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Abstract

The Crab Nebula (M1, NGC 1952), the remnant of the supernova observed in 1054 AD, is powered by a central pulsar (neutron star) spinning at 30.2 Hz with luminosity $\sim 5 \times 10^{31}$ W. The nebula expands at $\sim 1,500$ km/s across a ~ 11 -light-year diameter. This paper derives the Master Universal Gravity UQFF equation incorporating pulsar wind-driven expansion, magnetic field electron dynamics, cosmic expansion, and Aether electromagnetic correction. The result $g_{\text{Crab}} \approx 1.481 \times 10^6 \text{ m/s}^2$ is completely dominated by the relativistic pulsar wind term.

1. Introduction

Hubble's 2005 mosaic of the Crab Nebula (24 exposures) reveals intricate filaments of hydrogen, oxygen, and sulfur, plus wispy synchrotron features that evolve on timescales of days, driven by the pulsar's relativistic wind. The pulsar wind creates a shock front at ~ 0.1 pc, where electrons reach near-light speeds, producing synchrotron radiation. The nebula's magnetic field averages $\sim 10^{-8}$ T (stronger near the pulsar at $\sim 10^{-4}$ T). Under UQFF, the pulsar wind term completely dominates the gravitational evolution, with the Aether electromagnetic term providing a secondary non-standard correction.

2. Master UQFF Gravity Equation

$$\begin{aligned} & g_{\text{Crab}}(r, t) = (G M) / r^2 (1 + H(z)t) (1 + f_{\text{TRZ}}) \\ & + a_{\text{wind}} \\ & + M_{\text{mag}} \end{aligned}$$

Where:

- a_{wind} = pulsar wind-driven expansion acceleration
- M_{mag} = magnetic field electron dynamics term

2.1 Parameters

Parameter	Symbol	Value	Source
Nebula total mass	M	$4.6 \text{ MM}_{\text{sun}} = 9.149 \times 10^{30} \text{ kg}$	Hubble
Nebula radius	r	$5.2 \times 10^{16} \text{ m}$ (~5.5 ly)	Hubble
Pulsar luminosity	P_{pulsar}	$5 \times 10^{31} \text{ W}$	Labs
Expansion velocity	v_{shock}	$1.5 \times 10^6 \text{ m/s}$	Hubble
Filament density	ρ_{fil}	10-21 kg/m ³	Labs
Nebula B field	B	10-8 T (average)	Labs
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$	Standard
Redshift	z	0.0015	Distance calc
Time since SN	t	971 yr = $3.064 \times 10^{10} \text{ s}$	Historical
$\rho_{\text{vac,UA}}$	—	$7.09 \times 10^{-36} \text{ J/m}^3$	UQFF
$\rho_{\text{vac,SCM}}$	—	$7.09 \times 10^{-37} \text{ J/m}^3$	UQFF
f_{TRZ}	—	0.1	UQFF

3. Long-Form Derivation

Step 1: Base Gravitational Term
$$g_{\text{grav}} = (6.6743 \times 10^{-11} \times 9.149 \times 10^{30}) / (5.2 \times 10^{16})^2 \parallel \& = 6.106 \times 10^{-20} / 2.704 \times 10^{33} = 2.258 \times 10^{-13} \text{ m/s}^2 \end{aligned}$$

Step 2: Pulsar Wind Expansion Term
$$F_{\text{wind}} = (P_{\text{pulsar}} / (4\pi \times r^2)) \times (1 + v_{\text{shock}}/c) \parallel \& = (5 \times 10^{31} / (4 \times 3.1416 \times (5.2 \times 10^{16})^2)) \times (1 + 1.5 \times 10^6 / 3 \times 10^8) \parallel \& = (5 \times 10^{31} / 3.393 \times 10^{34}) \times 1.005 \parallel \& = 1.474 \times 10^{-3} \text{ N/m}^2 \parallel \& a_{\text{wind}} = 1.481 \times 10^{-3} / \rho_{\text{fil}} = 1.481 \times 10^{-3} / 1 \times 10^{-21} = 1.481 \times 10^{18} \text{ m/s}^2 \parallel \& \text{a}_{\text{wind}} = 1.481 \times 10^{18} \times 10^{-12} = 1.481 \times 10^6 \text{ m/s}^2 \end{aligned}$$

Step 3: Magnetic Field Electron Dynamics
$$q \times (v \times B) = 1.602 \times 10^{-19} \times 1.5 \times 10^6 \times 1 \times 10^{-8} = 2.403 \times 10^{-21} \text{ N} \parallel \& M_{\text{mag}} = 2.403 \times 10^{-21} / m_e = 2.403 \times 10^{-21} / 9.11 \times 10^{-31} = 2.638 \times 10^9 \text{ m/s}^2 \parallel \& M_{\text{mag}} = 2.638 \times 10^9 \times 10^{-12} = 2.638 \times 10^{-3} \text{ m/s}^2 \end{aligned}$$

Step 4: Cosmic Expansion
$$H(z) = 2.269 \times 10^{-18} \text{ s}^{-1} (z = 0.0015) \parallel \& H(z) \times t = 2.269 \times 10^{-18} \times 3.064 \times 10^{10} = 6.952 \times 10^{-8} \parallel \& 1 + H(z) \times t \approx 1.00000007 \end{aligned}$$

Step 5: Time-Reversal Correction
$$1 + f_{\text{TRZ}} = 1.1$$

Step 6: Final Solution
$$g_{\text{Crab}} = (2.258 \times 10^{-13}) \times (1.00000007) \times (1.1) + 1.481 \times 10^6 + 2.638 \times 10^{-3} \parallel \& = 2.484 \times 10^{-13} + 1.481 \times 10^6 + 2.638 \times 10^{-3} \parallel \& \approx 1.481 \times 10^6 \text{ m/s}^2 \end{aligned}$$

4. Physical Interpretation

The Crab Nebula is unique among UQFF systems: the pulsar wind term ($1.481 \times 10^6 \text{ m/s}^2$) exceeds all other terms by orders of magnitude. Classical gravity (2.258×10^{-13}) is negligible. The magnetic field electron dynamics term (2.638×10^{-3}) provides a secondary UQFF correction coupling the electron population to [SCm]. The cosmic expansion term is effectively zero over the 971-year age of the remnant — validating UQFF's correct near-zero cosmological behavior at short timescales.

5. UQFF Framework Advancement

- UQFF successfully handles pulsar-dominated supernova remnant physics
- Pulsar wind term expressed as radiation pressure / filament density $\times$ relativistic correction
- Electron-mass magnetic term (M_{mag}) demonstrates UQFF mass-scale flexibility
- Validates UQFF for extreme energy environments (pulsar wind nebulae)

6. Conclusions

The Master UQFF gravity equation for the Crab Nebula yields $g_{\text{Crab}} \approx 1.481 \times 10^6 \text{ m/s}^2$, completely dominated by the relativistic pulsar wind pressure term. This is the most extreme result in the batch, demonstrating UQFF's dynamic range from 10^{-3} m/s^2 (nebulae) to 10^6 m/s^2 (pulsar wind nebulae). The result confirms UQFF handles relativistic energy injection accurately while preserving all non-standard correction terms.

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<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi_i}}$ jet
> modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- L_{Edd} = $4\pi G M m_p c / \sigma_{\text{T}}$ is the classical Eddington luminosity
- ρ_{SCm} = $7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density

- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M– σ correction (PAPER_1048): The phonon-corrected M– σ relation becomes

$$M_{\text{BH}} \propto \sigma^{4+\delta} \text{ where } \delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$$

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

- > Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
- > PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
- > (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa_i, n/26}}\right]$$

where $(a)_n = a(a+1) \cdots (a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{4^{4n}} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa_i, n/26}}\right]$$

This sum converges absolutely for $|SSq| < 1$ (satisfied by $SSq = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $SSq \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_i, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function

$$Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$$

unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_{\mathrm{NS}})(\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,SCm}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac,SCm}} \cdot \phi_{\mathrm{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2) \phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b,seed}} \rightarrow \text{4 forces} \\ F_{U_{\mathrm{Bi}}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi_{\mathrm{NS}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,SCm}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,SCm}} \cdot \exp[-\exp(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}})]$$

For this system, the local VDS sub-ratio is 0.164 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m³.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 59, \quad n_{\mathrm{channel}} = 13/26$$

Since $p_{\mathrm{DVP}} = 59$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}}) = 1$

constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.164	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 59$	PASS Resonant
BSH layers	26 harmonic terms $j = 1\dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection	
κ decay	5.0×10^{-4} day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	PASS Consistent
Proton decay rate	$\kappa = 0.0005/\text{day}$	Γ_p suppression	< $4.17 \times 10^{-35}/\text{yr}$	Super-K 2024 PASS Consistent
UQFF buoyancy signature	$F_{U_{Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{U_{Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger F_U_Bi Strain Suppression & BCS Gap
PAPER_1011	GW170817 NS Merger F_U_Bi_i 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_U_Bi_i with S26(3)
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1047	Type Iax Supernova Buoyancy Reversal

15 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,Bi\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms

| s26_wstp_kernel.py | 8-symbol Wolfram export (UQFFS26) | S26, R26, NaiveLi, S26VDS |

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}}/(1-2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

| Module | Purpose | Key Result |

|-----|-----|-----|

| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

Core equations:

- f_26(q) = $\text{Sum}_{\{n=0\}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$

- phi_26(q) = $\text{Sum}_{\{n=0\}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$

- psi_26(q) = $\text{Sum}_{\{n=1\}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_pi_uqff.py | Classical + UQFF-modified 1/pi + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |

| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

Core equation: $1/\pi = (2\sqrt{2})/9801 \text{ Sum } R_n (1103+26390n) W_{26}(n) / C_{26}$

where $W_{26}(n) = \text{Prod}_{\{i=1\}^{\{26\}} [1 + [SSq]\exp(-kappa*i/n/26)]$

S204.5 Calibration Constants (Canonical)

| Symbol | Value | Description |

|-----|-----|-----|

| [SSq] | 0.57 | Universal Quantized Factor |

| kappa | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate |

| beta_i | 0.603 | Buoyancy coupling coefficient |

| H_SCm | 0.99 | SCm manifold completeness |

| rho_SCm | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density |

| rho_UA | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density |

| omega_SCm | $2*\pi \times 1.25 \text{ THz}$ | SCm phonon resonance |

| sigma_0 | 10^{-4} | Base neutron cross-section |

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

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